

Checkpoint Assessment – SQL Analytical Skills

Problem Statement: Sales Performance Analysis using Retail_Sales Database

Build a **SQL-based analytical report** using the **Retail_Sales database**.

The objective is to analyze sales performance, customer behavior, and product insights using **advanced SQL techniques**.

Objective

Use SQL queries to:

- Explore and analyze relational data
 - Apply **SQL functions and aggregations**
 - Use **joins to combine multiple tables**
 - Implement **subqueries and window functions**
 - Generate meaningful business insights from transactional data
-

Time Allotment

2 Hours

Prerequisites

Learners must have knowledge of:

- SQL SELECT statements
 - Filtering using **WHERE**
 - Aggregate functions (**SUM, AVG, COUNT, MIN, MAX**)
 - **GROUP BY** and **HAVING**
 - SQL **JOIN** operations
 - Subqueries
 - Window Functions
-

Dataset Link : [Retail_Sales](#)

Dataset

The **Retail_Sales database** includes the following key tables:

Table	Description
Customer	Customer information
Employee	Employee information
Product	Product details
Category	Product categories
Supplier	Supplier details

SalesOrder	Customer orders
OrderDetail	Order line items
Shipper	Shipping companies

Functional Requirements

1. Basic Data Exploration

Write SQL queries to:

1. Retrieve all records from the **Product** table.
2. Display product names and unit prices for products costing more than **\$50**.
3. List all customers located in **Germany**.

2. SQL Functions & Aggregations

Use SQL aggregate functions to answer the following:

1. Calculate the **total number of orders** placed.
2. Find the **average unit price of all products**.
3. Determine the **total revenue generated**.

[Hint: Revenue = **UnitPrice** × **Quantity** × (**1 - Discount**)]

3. Join Operations

Use joins to combine tables and answer the following:

a) Product Category Analysis

Display:

- Product Name
- Category Name
- Unit Price

Tables involved:

- Product
- Category

b) Customer Order Analysis

Display:

- Customer Company Name
- Order ID
- Order Date

Tables involved:

- Customer
- SalesOrder

c) Sales Performance by Employee

Display:

- Employee Name
- Order ID
- Customer Name

Tables involved:

- Employee
- SalesOrder
- Customer

4. Subqueries

Use subqueries to answer the following:

1. Find products whose **unit price is higher than the average price of all products**.
2. Retrieve customers who have placed **more than 5 orders**.
3. Identify products that **have never been ordered**.

5. Window Functions

Use window functions to perform analytical queries.

a) Rank Products by Price

Rank products based on their **unit price within each category**.

b) Running Total of Sales

Calculate the running total of revenue by order date.

Expected columns:

- Order Date
- Order Revenue
- Running Total Revenue

c) Top Customers by Revenue

Calculate total revenue generated by each customer and assign a rank.

Expected output:

- Customer Name
- Total Revenue

- Customer Rank

6. Advanced Analytical Queries

Write SQL queries to answer the following business questions:

1. Which product category generates the highest revenue?
 2. Which customer has placed the most orders?
 3. Which employee handles the highest number of orders?
-

Technical Constraints

- Use MySQL / MySQL Workbench
 - Queries must be written manually
 - Do not modify database schema
 - Avoid using external tools
-

Submission Guidelines

Students must submit:

1. SQL Script File (.sql) containing:
 - All queries
 - Proper comments explaining each query
2. Query Results Screenshot or Export
3. Ensure queries run without errors.

